queue until the message is retrieved by the destined server. A few open source choices of message queues are RabbitMQ, Apache Kafka, and Apache ActiveMQ.

Virtualization

When we think of virtualization, we think of a methodology that separates our resources into different environments. The latter can have different execution environments based on the resources allocated by the users. These resources can be based on computing power, memory, disk space, and network. In virtualization, we are asking the existing interface to mimic the behaviour of another system.

Virtualization can help its users diversify the development and production environment. Based on the application, the user can decide on the operating system flavour and the libraries. Virtualization enables a high degree of flexibility and portability. Many tech giants like Amazon Web Services (AWS) and Azure offer virtual machines (VMs), and users can host their applications on these VMs. These modern-day distributed systems are called data centers and clouds. The infrastructure difference between the deployment of regular and virtualized applications can be seen in Figure 7.

Virtualization can be categorised into three types:

- Hardware virtualization (Type-1)
- Desktop virtualization (Type-2)
- Containerization (Type-3)

Hardware virtualization: Suppose you want to run Ubuntu on your Windows machine — sounds tricky, right? Well, it's possible with hardware virtualization. In this sort of virtualization, we have a guest machine over the host machine. These terminologies — host and guest — are used to define a physical machine and a virtual machine, respectively. This separation of the guest machine from the host machine is done via software or firmware, known as a hypervisor.

Desktop virtualization: For hardware virtualization, we were hosting a guest machine over the host machine. But what if we want to separate the logical desktop from the hardware itself? This is where desktop virtualization comes in. When users want to interact with such a desktop setup via desktop virtualization, they first connect to a network and then use a keyboard, mouse or touchpad to connect to it. The virtualized desktop can be accessed via another machine or mobile device. The host machine in this scenario can cater to multiple virtual machines at the same instance for multiple users. With desktop virtualization, an organisation can enjoy perks like scalability and a reduction in capital expenditure.

Containerization: Containerization can be classified as OS level virtualization. It is a feature of the modern-day operating system where the kernel allows the existence of multiple isolated user-space instances of these containers. All the requests by these containers are then forwarded to the host machine's kernel to entertain them. For the user the container may look like a real computer, but it is just another process for the host machine. The best thing about being isolated in the container is that any application or service running on it can only see the container's contents and its attached devices assigned by the host machine. A few important containers are:

- Docker
- AWS Fargate
- Canonical LXC
- OpenVZ
- CoreOS

In the next part of this series of articles, we will discuss containerization in detail, including container architecture, container engine, container networking, and container runtime. Till then keep exploring virtualization! END

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